QUANTUM MECHANICS

I. HEISENBERG UNCERTAINTY PRINCIPLE

This principle is a direct consequence of the dual nature of matter, and is also known as '*The Principle of indeterminacy*'.

Statement:

"*In any simultaneous determination of the position and momentum of the particle, the product of the uncertainty in the x- coordinate of the particle position (Δx) in motion and the uncertainty in the x- component of the momentum (Δp_x) is of the order of or greater than h/4π*".

$$\boxed{\Delta x.\Delta p_x \geq \frac{h}{4\pi}} \rightarrow (1)$$

Where Δ → minimum uncertainty.

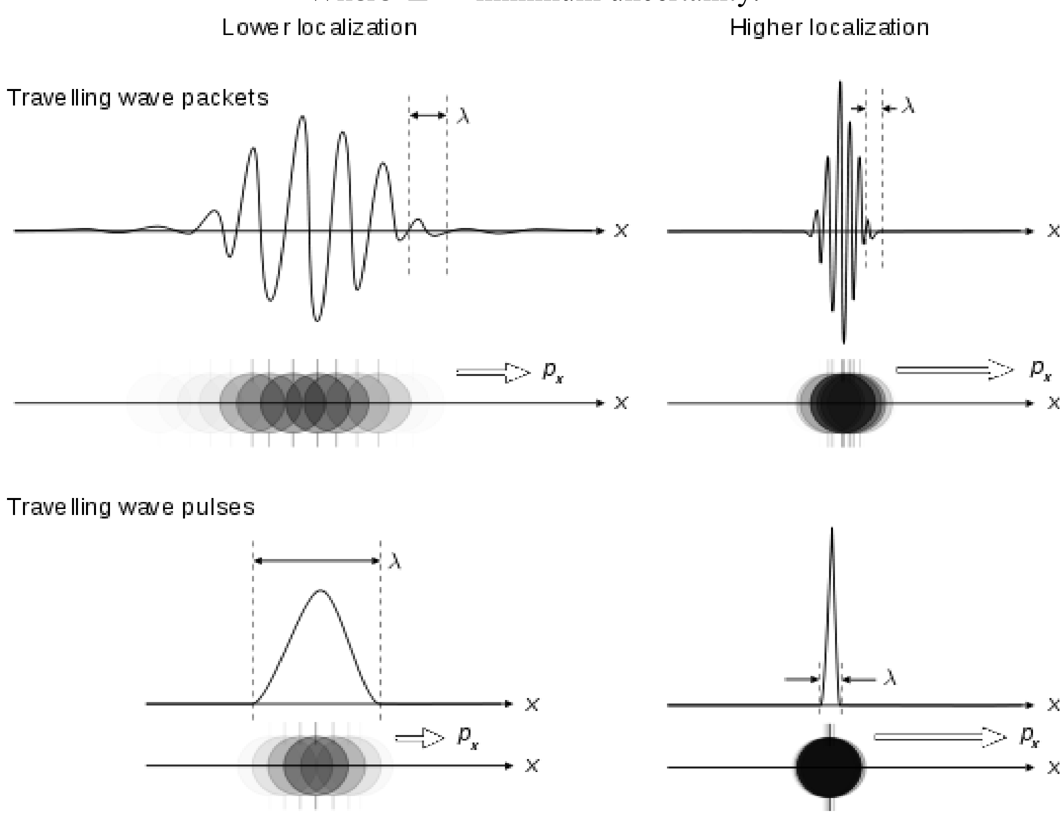

Figure – 1

Explanation:

According to the principle, '*If the momentum of the particle (ex: electron) is known then by the relation λ = h/p the wavelength of the associated wave has a unique value i.e. wave is monochromatic. In such cases, the wave packet will have the infinite width and the velocity spread is small so that particle velocity can be determined accurately but the position of the particle may be anywhere between $+\infty$ to $-\infty$ i.e., particle position is uncertain. If the position of the particle is fixed by making the wave packet very small then the wavelength will be too* *small so that the velocity spread is very large hence by $\lambda = h/p$ the momentum will become uncertain*'.

Based on these observations, Heisenberg concluded that – It is physically impossible to determine precisely and simultaneously the position and momentum of the particle.

Other forms:

The Heisenberg uncertainty principle can also be expressed in terms of the uncertainties involved in the measurements of two more conjugate pairs (canonical):

(i) Energy (E) and time(t) as $\boxed{\Delta E.\Delta t \geq \frac{h}{4\pi}}$ → (2)

(ii) Angular momentum (L) and angular displacement (θ) as $\boxed{\Delta L.\Delta\theta \geq \frac{h}{4\pi}}$ → (3)

Physical Significance:

In Classical mechanics, every particle can be assigned by a definite location in space and possesses a definite momentum at an instant of time and also at any later point of time i.e. *the trajectory of the particle can be traced precisely*.

But in quantum mechanics, this concept breaks down in the atomic scale because the quantum nature of interactions on the atomic scale introduces uncontrollable disturbance in the system under measurements and hence there will be uncertainties in the simultaneous measurement of position and momentum. Also, this uncertainty is not due to the defect in the method of measurement involved for momentum and position, but they *'arise due to the inherent limitations of nature'*. However precise may be the method of the measurement there is no escape from the uncertainty.

If the position coordinate 'x' of a particle in motion is determined at some instant so that $\Delta x = 0$, then at the same instant $\Delta p_x = \infty$. Similarly, if $\Delta p_x = 0$, then $\Delta x = \infty$. Thus the pair Δx & Δp_x is said to be conjugate and there should be no more accuracy but the degree of accuracy.

Thus Heisenberg Uncertainty principle signifies that –'*one should not think of the exact position or an accurate value for a momentum, instead we should think of the probable value for momentum and probability for finding the particle at a certain position. These probabilities in quantum mechanics are referred to as probability density functions'*.

II. WAVE FUNCTION

Definition:

Waves, in general, are associated with quantities that vary periodically. For example, sound waves are characterized by the pressure variation, water waves are characterized by the periodic variations of the height of the water surface at a point, etc., In the case of matter waves, the variable quantity is the wave function.

Therefore, "*The periodic variable quantity characterizing the matter waves is called wave function*". The wave function is always denoted by 'Ψ' (psi). The value of the wave function associated with a moving body like electron at a particular point x, y, z in space at the time 't' is denoted by:

$$\text{“}\psi(x,y,z,t)\text{”}$$

Where, x,y,z → are called position co-ordinates (or) space co-ordinates.
t → time co-ordinate.

Therefore, '*Ψ (x,y,z,t) is related to the likelihood of finding the particle at that point at that time'*.
The wave function is always a single-valued function of position and time.

Explanation:

The wave function in quantum mechanics accounts for the wave-like properties of a particle. It is obtained by solving a second-order differential equation called "*Schrodinger's wave equation*". The Schrodinger wave equation plays the role in quantum mechanics as, like Newton's laws and conservation of energy in classical mechanics i.e., *it predicts the future behavior of a dynamic system*. This Schrodinger's wave equation can be set up in two contexts:

(i) *Time-dependent Schrodinger's wave equation*: In this case, the equation takes care of both the position and time variations of the wave function. It involves an imaginary quantity 'i' as:

$$\frac{-h^2}{8\pi^2 m}\frac{d^2\psi}{dx^2}+\psi.V=\frac{ih}{2\pi}\frac{d\psi}{dt} \qquad \rightarrow (4)$$

(ii) *Time -independent Schrodinger's wave equation*: In this case, the wave function can have variations only with the position but not with time i.e., steady states as:

$$\frac{d^2\psi}{dx^2}+\frac{8\pi^2 m}{h^2}(E-V).\psi=0 \qquad \rightarrow (5)$$

It is postulated in quantum mechanics that the solution to Schrödinger's equation will be associated with the number density of non-zero rest mass particles (photons). The wave function obtained as the solutions of the time-dependent Schrodinger's wave equation will always be complex. But the wave function for time-independent Schrodinger's wave equation need not necessarily be complex but for certain conditions, it will be a complex function. Thus a direct measurement of complex quantity (wave function) is not possible by any physical instrument i.e., *the Wave function is not an observable quantity and hence the wave function has no direct physical significance*.

Significance of Wave function: (Probability Density)

Since the wave function itself has no direct physical significance, and also because it is a complex number, we can think of the probability of finding the particle. i.e., something is somewhere at a given time. And as per the quantum mechanical postulate, the probability can have a value between 0 and 1.

Where (i) '0' corresponds to the certainty of absence.
(ii)'1' corresponds to the certainty of presence. (It is 1 because a single particle cannot be more than at one place at any instant.)

Thus, the probability of finding the particle must be *'positive and real'*. Since the wave function is complex, consider the complex conjugate of the complex number and take the product of those two. Thus, the product of $\psi(x,y,z,t)$ and $\psi^*(x,y,z,t)$ is real and positive if $\psi(x,y,z,t) \neq 0$. The fact expressed can be further interpreted as:

$$\boxed{\psi(x,y,z,t).\psi^*(x,y,z,t) = |\psi\ (x,y,z,t)|^2} \qquad \rightarrow (6)$$

The above is called '*Max Born Interpretation*' according to which the square of the magnitude of the wave function evaluated at a particular point represents the probability of finding the particle at that point. Thus equation (6) is the probability density P (x, y, z) {i.e., *probability per unit volume*} for the motion of the particle in three dimensions at positions (x,y,z) at the time 't'.
If we define a small volume element dτ = dx.dy.dz, then the probability of finding the particle in a volume element will be:

$$\boxed{P(x,y,z,t).d\tau = \psi(x,y,z,t).\psi^*(x,y,z,t).d\tau = |\psi\ (x,y,z,t)|^2.d\tau} \qquad \rightarrow (7)$$

Where, P(x,y,z) .dτ → *Probability per unit volume.*
Similarly, the probability of finding the particle on a plane is given by:

$$\boxed{P(x,y).dx.dy = \psi(x,y).\psi^*(x,y).dx.dy = |\psi\ (x,y)|^2.dx.dy} \qquad \rightarrow (8)$$

Where, P(x,y) .dx.dy → *Probability per unit area.*
and, Probability of finding the particle along the line is given by:

$$\boxed{P(x).dx = \psi(x).\psi^*(x).dx = |\psi\ (x)|^2.dx} \qquad \rightarrow (9)$$

Where, P(x) .dx → *Probability per unit length.*

Note: 'Ψ (x,y,z,t)', 'Ψ ($\vec{r}$,t)' and simply 'Ψ' all are the same.

Normalization of the Wave function:

If '$\psi(x,y,z,t)$' is the wave function of a particle which represents the *probability amplitude*, then $|\psi\ (x,y,z,t)|^2.d\tau$ is the *probability density*. Where '$d\tau = dx.dy.dz$' is the small volume element inside the volume 'τ'. (Figure-2)

Here, the total probability of finding the particle in the volume element dτ must be:

$$\boxed{\int_0^\tau |\psi(x,y,z,t)|^2.d\tau = 1 \ (\text{or})\ \int_0^\tau |\psi|^2.d\tau = 1} \qquad \rightarrow (10)$$

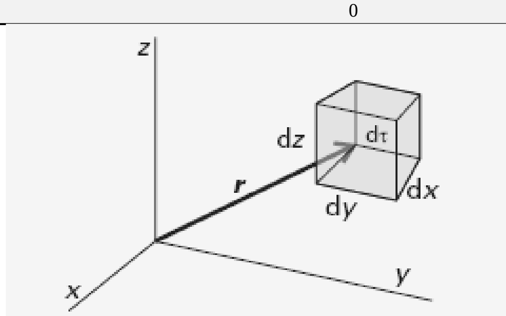

Figure – 2

Therefore, the total probability of finding the particle somewhere in space must be equal to unity. Hence Ψ (x,y,z,t) must satisfy the condition:

$$\int_{-\infty}^{+\infty} |\psi(x,y,z,t)|^2 .d\tau = 1 \quad \text{(or)} \quad \int_{-\infty}^{+\infty} |\psi|^2 .d\tau = 1 \qquad \rightarrow (11)$$

The wave function which satisfies the above condition is said to be *"normalized wave function" (or) "normalized to unity"*. Every acceptable wave function must be *normalizable*.

Orthogonality of Wave function:

If for a given system, there are two different wave functions say ψ_i and ψ_j and if they can be normalized then:

$$\int \psi_i \psi_i^* .d\tau = 1$$
$$\int \psi_j \psi_j^* .d\tau = 1$$

Now, the wave functions ψ_i and ψ_j are said to be *mutually orthogonal wave function* if they satisfy the following conditions:

$$\int \psi_i \psi_j^* .d\tau = 0 \quad \text{for } i \neq j \qquad \rightarrow (12)$$
$$\int \psi_j \psi_i^* .d\tau = 0 \quad \text{for } i \neq j$$

The wave functions which are orthogonal and also normalized are called *"Orthonormal Functions"*.

Properties (or) Limitations of the Wave function:

1. The wave function must be finite for all values of x, y, z.
2. The wave function must be single-valued i.e., for every set of values of (x, y, z) ψ must have only one value.
3. The wave function must be continuous in all regions except in those regions where the potential energy V(x,y,z) = ∞ .
4. The partial derivatives of ψ i.e. $\frac{\partial\psi}{\partial x}, \frac{\partial\psi}{\partial y}, \frac{\partial\psi}{\partial z}$ must be continuous, single-valued and finite everywhere.
5. The wave function vanishes at the boundaries.

The wave function which satisfies these above conditions is called a "*well-behaved function*".

III. TIME –INDEPENDENT SCHRODINGER WAVE EQUATION

Schrodinger wave equation is an equation capable of determining the wave function of the matter waves in any different physical situations.

Let a particle of mass 'm' be in motion along the positive x-axis with accurately known momentum 'p' and total energy 'E'. The position of such a particle would be completely undetermined.

According to the de-Broglie theory, the wave associated with such particle should be a plane continuous harmonic wave of wavelength: $\lambda = h/p$. Where, $p = mv$.

Let us consider the wave function for a de-Broglie wave traveling in x-direction be in complex notation as $\psi(x) = A.e^{i(kx-\omega t)}$ → (13)

Where, A → amplitude of wave (constant), ω → angular frequency of the wave,
k → propagation constant.

Differentiating equation (13) twice w.r.t.'t' we get:

$$\frac{d^2\psi}{dt^2} = -\omega^2\left[A.e^{i(kx-\omega t)}\right] \text{ (or) } \frac{d^2\psi}{dt^2} = -\omega^2\psi \quad \rightarrow (14)$$

We have the equation for the traveling wave as:

$$\frac{d^2y}{dx^2} = \frac{1}{v^2}\frac{d^2y}{dt^2}$$

Where, y → displacement of the particle
v→ velocity of the wave.

By analogy, we can write the wave equation for the motion of the particle as

$$\frac{d^2\psi}{dx^2} = \frac{1}{v^2}\frac{d^2\psi}{dt^2} \quad \rightarrow (15)$$

From the equations (14) and (15) we get: $\frac{d^2\psi}{dx^2}v^2 = -\omega^2\psi$ (or) $\frac{d^2\psi}{dx^2} = -\frac{\omega^2}{v^2}\psi$

But, ω = 2πγ and v = γλ (Where, λ → wavelength of the wave and γ → frequency of the wave)

On substituting we get: $\frac{d^2\psi}{dx^2} = -\frac{4\pi^2\omega^2}{\gamma^2\lambda^2}\psi$ (or) $\frac{1}{\lambda^2} = -\frac{1}{4\pi^2\psi}\frac{d^2\psi}{dx^2}$ → (16)

If a particle is acted upon a force field from a potential 'V' then its total energy is given by:

$$E = P.E + K.E$$

$$E = V + K.E \quad \rightarrow (17)$$

But kinetic energy is: $K.E = \frac{1}{2}mv^2 = \frac{m^2v^2}{2m} = \frac{p^2}{2m} = \frac{\left(h/\lambda\right)^2}{2m}$

$$K.E = \frac{h^2}{2m}\left(\frac{1}{\lambda^2}\right) \text{ Or } K.E = \frac{-h^2}{8\pi^2 m}\frac{1}{\psi}\frac{d^2\psi}{dx^2}$$

Now equation (17) becomes: $E = \frac{-h^2}{8\pi^2 m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V$

By re-arranging: $\boxed{\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-V).\psi = 0}$

This equation is called the *time-independent Schrodinger wave equation* for a particle moving in one dimension.

In three dimension it is: $\boxed{\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2 m}{h^2}(E-V).\psi = 0}$ → (18)

IV. EIGEN VALUES AND EIGEN FUNCTION

For bound states i.e., for particles confined to a finite region and for single-valued & continuous the Schrodinger equation is given by: $\frac{d^2\psi}{dx^2}+\frac{8\pi^2m}{h^2}(E-V).\psi=0$

For the above equation there can only be certain definite values of energy 'E' with boundary conditions, these values of E are the allowed values of E of a particular system and are said to be '*Characteristic values' or 'Proper values or 'Eigenvalues'*. Corresponding to each allowed values of E there is one function 'ψ' obtained as solution of the above equation, this function is referred to as '*Characteristic function' or 'Wave function' or 'Eigenfunction'*.

V. APPLICATION OF SCHRODINGER WAVE EQUATION:

1 .Energy Eigenvalues and Eigenfunctions for a particle in a one dimensional potential well of infinite height : (Particle in a box).

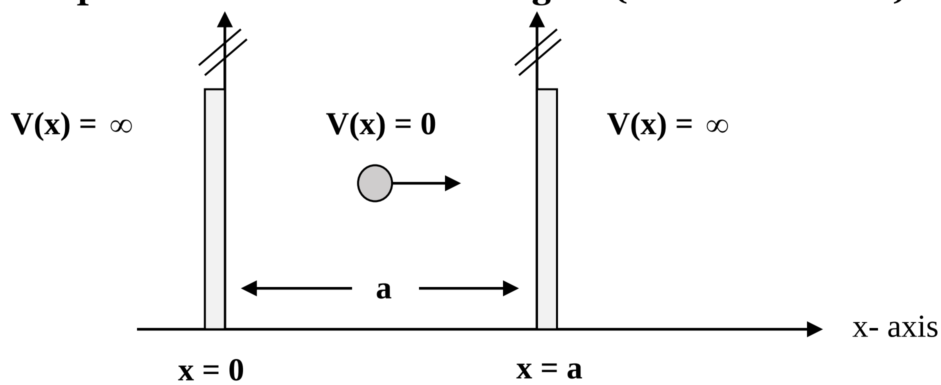

Figure – 3

Consider a particle of mass 'm' confined within a region of 0 to a on the x-axis in a box.
At x = 0 and x = a there are two rigid impenetrable potential walls of infinite height.
Then: (i) Potential energy $V(x) = \infty$, for $x \leq 0$ and $x \geq a$.
(ii) Potential energy $V(x) = 0$, for $0 < x < a$. ($\therefore$ no force acts on particle.)
Note that when the particle collides with rigid walls there is no loss of energy so that the total energy 'E' of the particle remains constant.
Now, for a particle outside the potential box the Schrodinger wave equation can be written as:

$$\frac{d^2\psi}{dx^2}+\frac{8\pi^2m}{h^2}(E-\infty)\psi=0 \quad \{\because\ V(x)=\infty\}$$

The above equation holds good if $\psi = 0$, which means that the particle cannot be found outside the box.
For a particle outside the potential box the Schrodinger wave equation can be written as:

$$\frac{d^2\psi}{dx^2}+\frac{8\pi^2m}{h^2}(E-0)\psi=0 \quad \{\because\ V(x)=0\}$$

$$\text{Put, } k^2 = \frac{8\pi^2 mE}{h^2} \qquad \rightarrow (19)$$

$$\therefore \frac{d^2\psi}{dx^2} + k^2\psi = 0 \qquad \rightarrow (20)$$

The above equation (26) is a second-order differential equation.

From figure-3, since the particle is moving in both '+x' and '-x' axis it is clear that we have two sinusoidal waves moving in the opposite direction. Thus, the solution to Schrodinger's equation inside the well is the solution of sinusoidal waves which are traveling in opposite directions and interfering with each other. Therefore let the solution be of the form:

$$\psi(x) = A.\cos(kx) + B.\sin(kx) \qquad \rightarrow (21)$$

Where A and B are constants depending on the boundary conditions.

We have the boundary conditions as:

(i) At x = 0, $\psi(x) = 0$. Then equation (21) becomes:

$$0 = A.\cos(0) + B.\sin(0)$$

$$\Rightarrow A = 0$$

(ii) At x = a, $\psi(x) = 0$. Then equation (21) becomes:

$$0 = A.\cos(ka) + B.\sin(ka)$$

$$0 = B.\sin(ka) \quad (\because A = 0)$$

Now $B \neq 0$ because there will be no solution. Therefore: $\sin(ka) = 0$

$$\Rightarrow ka = n\pi \ (\text{or}) \ \boxed{k = \frac{n\pi}{a}} \qquad \rightarrow (22)$$

Where, n = 1, 2, 3, 4.. is called Principal quantum number. (Here $n \neq 0$ because if $n = 0 \Rightarrow k = 0$ $\Rightarrow E = 0$ hence $\psi(x) = 0$ everywhere in the box i.e., particle with zero energy cannot be present inside the box). From equation (19) and (22), the energy of the particle can be written as:

$$E_n = \frac{k^2h^2}{8\pi^2 m} = \frac{\left(\frac{n\pi}{a}\right)^2 h^2}{8\pi^2 m}$$

$$\Rightarrow \boxed{E_n = \frac{n^2h^2}{8ma^2}} \qquad \rightarrow (23)$$

Where n = 1, 2, 3,….

In the above equation, the possible values of energy E are called *Eigenvalues* for the particle in one dimensional potential well of infinite height. This equation shows that the Eigenvalues of the energy are *discrete*.

From equation (23) we can conclude that:

1. For n = 1 $\Rightarrow E_1 = \frac{h^2}{8ma^2}$ this is called *ground state energy or zero-point energy*.

2. For $n > 1 \Rightarrow E_2 = \frac{4h^2}{8ma^2}, E_3 = \frac{9h^2}{8ma^2}$ …. These are called *excited state energies*.

Now the equation (3) becomes: $\psi_n(x) = B.\sin\left(\frac{n\pi x}{a}\right)$ $\rightarrow (24)$

Since B is undetermined the equation (24) cannot represent the Eigen function.

Evaluation of B (Normalization): According to the normalization condition, the total probability of the particle in the box of width 'a' must be unity i.e.,

$$\int_0^a \psi\psi^* dx = 1 \quad \text{or} \quad \int_0^a |\psi_n(x)|^2 dx = 1$$

$$\text{But, } \psi_n(x) = B.\sin\left(\frac{n\pi x}{a}\right)$$

$$\therefore \int_0^a \left|B\,.\,\sin\left(\frac{n\pi x}{a}\right)\right|^2 dx = 1 \quad \Rightarrow \quad B^2\int_0^a \sin^2\left(\frac{n\pi x}{a}\right)dx = 1$$

$$\text{But, } \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$\therefore \frac{B^2}{2}\int_0^a\left(1-\cos\left(\frac{2n\pi x}{a}\right)\right)dx = 1 \quad \Rightarrow \quad \frac{B^2}{2}\left[x - \frac{a}{2n\pi}\sin\left(\frac{2n\pi x}{a}\right)\right]_0^a = 1$$

For both the limits the second term of the integrated expression becomes zero, therefore:

$$\frac{B^2 a}{2} = 1 \quad \text{(or)} \quad B = \sqrt{\frac{2}{a}}$$ $\rightarrow$This is called the '*normalization constant*'.

$$\boxed{\therefore \psi_n(x) = \sqrt{\frac{2}{a}}.\sin\left(\frac{n\pi x}{a}\right)} \rightarrow (25)$$

Where n =1, 2, 3, 4….

The possible values of the above-normalized wave function $\psi_n(x)$ are called '*Eigenfunction'* for a particle in one dimensional potential well of infinite height.

VI. REPRESENTATION OF WAVE FUNCTION FOR A PARTICLE IN AN ONE DIMENSIONAL POTENTIAL WELL

For a particle in one dimensional potential well of infinite height we have:

Eigenfunction as**:** $E_n = \frac{n^2h^2}{8ma^2}$ And, Eigenfunction is: $\psi_n(x) = \sqrt{\frac{2}{a}}.\sin\left(\frac{n\pi x}{a}\right)$

Then the probability density for one dimension motion is:

$$P_n(x) = |\psi_n(x)|^2.dx = \frac{2}{a}\sin^2\left(\frac{n\pi x}{a}\right)$$

Now the probability of finding the particle along the x-axis is maximum when:

$$\frac{n\pi x}{a} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots \text{ (or) } x = \frac{a}{2n}, \frac{3a}{2n}, \frac{5a}{2n}, \ldots$$

Now the above variables can be represented as below in figure-4:

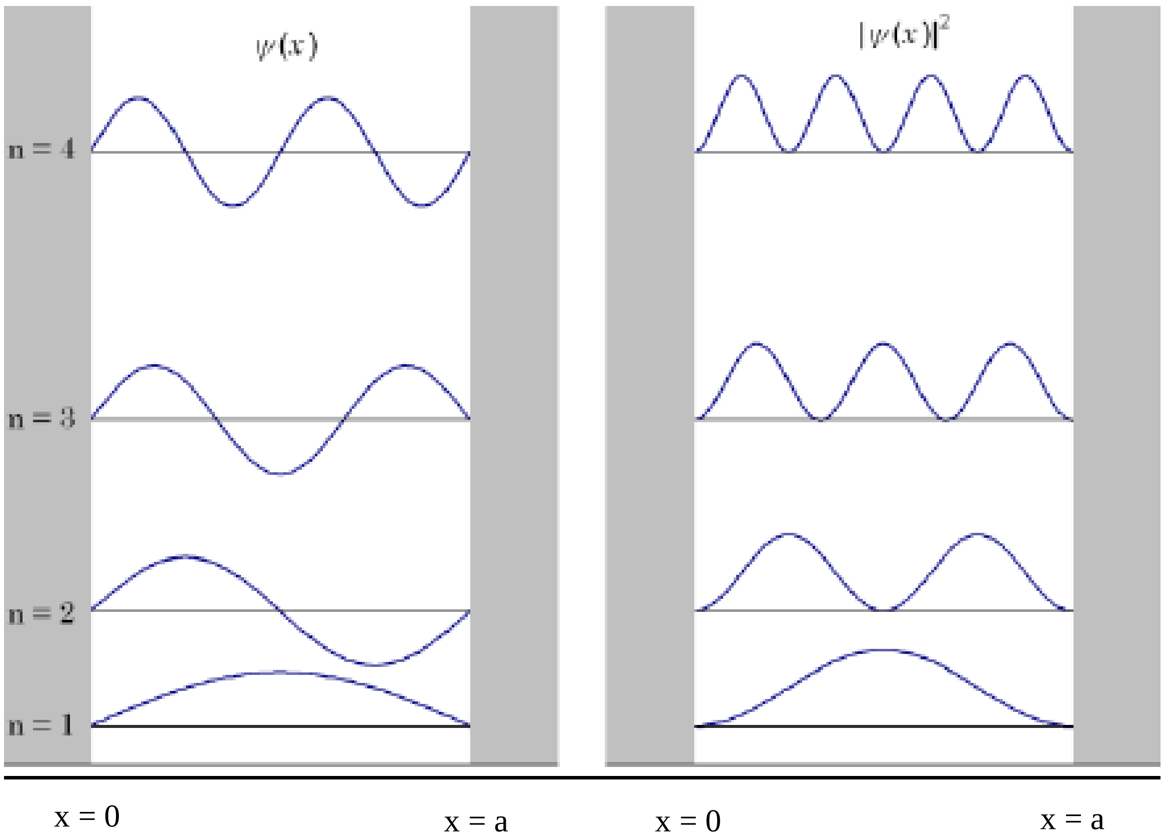

Figure-4

VII. EXPECTATION VALUES:

The expectation or mean value of an observable is a concept taken more or less directly from the classical theory of probability and statistics. It represents the (probability-weighted) average value of the results of many repetitions of the measurement of an observable, with each measurement carried out on one of very many identically prepared copies of the same system. The standard deviation of these results is then a measure of the spread of the results around the mean value and is known, in a quantum mechanical context, as the uncertainty.

Ex: If a state of a particle is known by a wave function $\psi(x.t)$ then, position operator is given by:

$$\langle x \rangle = \int_{-\infty}^{+\infty} x.\left|\psi(x,t)\right|^2 .dx$$

i.e., <x> is the average of measurements performed on particles all in the state ψ, which means that either you must find someway of returning the particle to its original state after each measurement, or else you prepare a whole ensemble of particles, each in the same state ψ, and measure th epositions of all of them: <x> is the average of these results.

Hence, the momentum expectation would be : $\langle p \rangle = \int_{-\infty}^{+\infty} \psi^*(x,t)\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)\psi(x,t).dx$

This suggests that the momentum be represented by the differential operator: $-i\hbar\frac{\partial}{\partial x}$

VIII. OPERATORS:

An operator is an 'mathematical object' that maps one state vector into another. In a physical system, there is a quantum mechanical operator that is associated with each measurable parameter. Generally an operator is anything that is capable of doing something to a function, hence there is an operator corresponding to every observable quantity. But, in quantum mechanics we deal with wave or wavefunction, thus when an operator operates on a wavefunction it must give the observable quantity times the wavefunction.

i.e., If we consider an operator represented by A corresponding to the observable quantity 'a', then:

$$A\psi = a\psi$$

Wavefunction that satisfies the above condition is called *eigen function* and the corresponding observable quantity is called *eigen values* and the equation is called *eigen value equation*.

Classical Quantity	Quantum Mechanical Operator
Position x, y, z	$\hat{x}, \hat{y}, \hat{z}$
Momentum p	$-i\hbar\vec{\nabla}$
Momentum components p_x, p_y, p_z	$-i\hbar\frac{\partial}{\partial x}, -i\hbar\frac{\partial}{\partial y}, -i\hbar\frac{\partial}{\partial z}$
Energy E	$-i\hbar\frac{\partial}{\partial t}$
Hamiltonian (Time independent)	$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+V(r)$
Kinetic Energy	$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial r^2}$